



THE HONG KONG
UNIVERSITY OF SCIENCE
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SCHOOL OF
ENGINEERING

Week4: Machine learning

A3: An Ergonomic Risk Assessment Tool for Construction Workers with Smartphone in Pocket

In-class Tutorial

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0. Objectives

Description

Assignment A2: Build ML models using provided dataset, compare KNN/SVM/random forest models, and give improvement proposal.

Today's Learning Goals

By the end of this session, you should be able to:

- Identify the main motion sensors available in a smartphone
- Explain how sensor signals can be used to infer body motion and posture
- Collect and export smartphone motion data using Sensor Logger
- Understand how raw time-series data become machine-learning features
- Use a prepared dataset to classify standing, walking, bending, and squatting

1. Introduction

- **WMSDs symptoms**

Work-related musculoskeletal disorders (WMSDs) are a group of conditions affecting muscles, tendons, joints, nerves, and spinal structures, caused or worsened by workplace activities. These disorders often manifest through chronic pain, stiffness, and functional limitations, with significant impacts on workers' health and productivity.

Most prevalence construction WMSDs symptoms in South China (Lee, 2023):

- Lower back (26.0%),
- Neck (24.7),
- Shoulder (22.1%),

WMSDs

WMSDs are workplace injuries to the soft tissues such as:

- muscles
- tendons
- ligaments
- joints
- blood vessels
- nerves



Work-related Musculoskeletal Disorders (WMSDs) Causing Financial and Occupational Health Problems in Construction Industry

2. Scientific background - Smartphone sensors



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**How Does
Your Phone
Sense Orientation?**

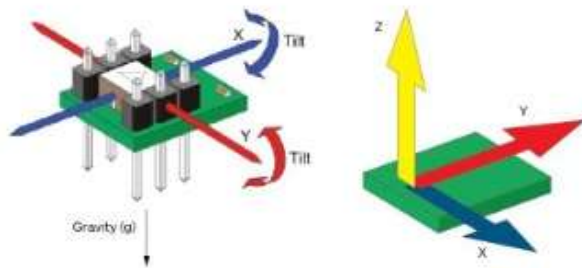


2. Scientific background

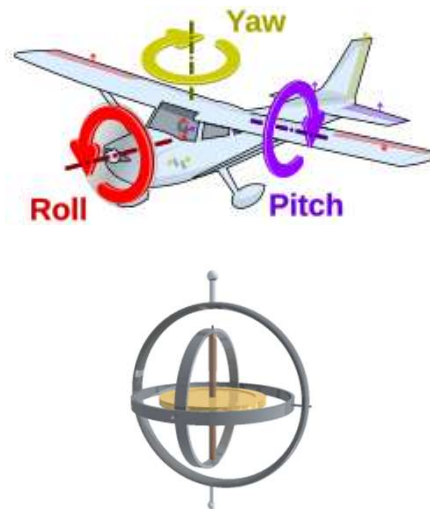
- How does your smartphone know its orientation?

<https://www.youtube.com/watch?v=pw-b8IGhb9o> (1:01-5:02)

Accelerometer



Gyroscope



Magnetometer



By combining the strengths of each sensor, the phone create a more stable estimation of its orientation.

2. Scientific background

- What does our dataset look like?



Ground truth video

Subject	Accelerometer_x(y,z)	Gyroscope_x(y,z)	Magnetometer_x(y,z)	seconds_elapsed	timestamp	Label_2
name	1.15648675	0.047643632	-18.89904022	146.0473408	1632193891330340000	2
name	1.022649884	-0.060479425	-17.6236496	146.0671755	1632193891350170000	2

Labelled dataset

Label_2: 1- standing, 2- walking, 5-bending, 8-squatting

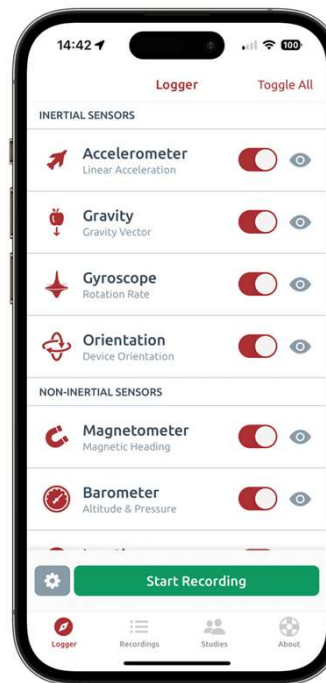
2. Scientific background

- **How to collect data?**

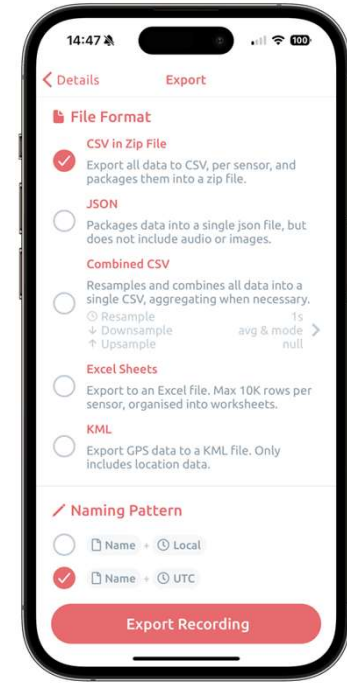
- Download *Sensor Logger* in App Store or Google Play (<https://www.tszheichoi.com/sensorlogger>)
- Select Accelerometer, Gravity, Gyroscope, Magnetometer and other supplement sensor signals to collect.
- Put the app installed smartphone into a belt pocket fixed around the worker's upper shirt pocket position.
- Start recording on both camera and Sensor Logger.
- Perform the work for at least 15 mins.
- End recording. Export the sensor data into csv zip files for later analyzation.



Experiment set up



Data collection



Result output

2. Scientific background

- **In-Class Activity: Collect your own data**

1. Download Sensor Logger from AppStore/Google Play on your phone.
2. Record your device IDs, placement locations.
3. Experiment Set Up
4. Start a short test recording and verify that the IMU data, timestamps, video, and device identifiers are being recorded correctly. (Optional)
5. Start recording and candidate follow the video for 3 mins.
6. Synchronize several times during the recording.
7. End recording and check that all files have been saved correctly

Experiment set up

1. Put smartphone in the upper pocket of the candidate.
2. Set up the camera with correct angle that covers all moving area of the candidate.
3. Synchronize Unix Timestamp for later labeling. (A Unix timestamp is the number of milliseconds elapsed since 1 January 1970 UTC, and in labelling it lets you align each sensor sample with the corresponding video time interval and assign the correct activity or posture label. For example, 1789395000000)

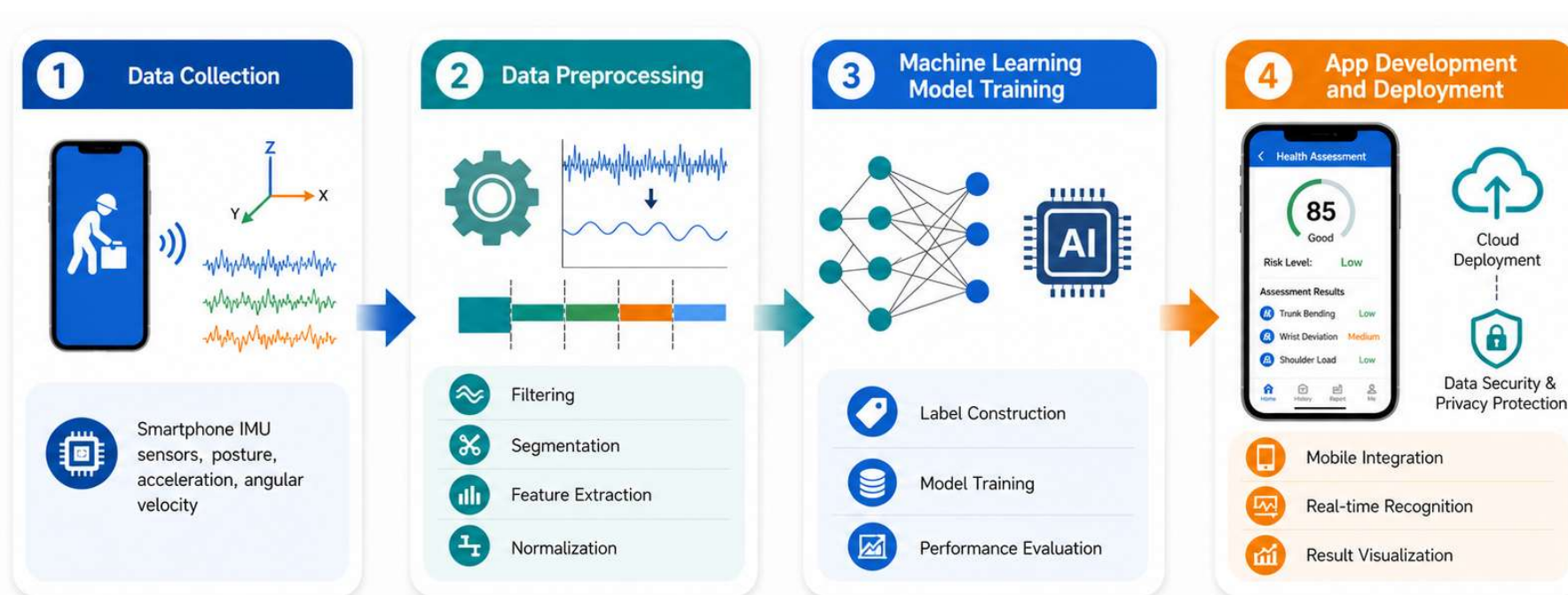
In-Class Activity:

Data Collection Demonstration



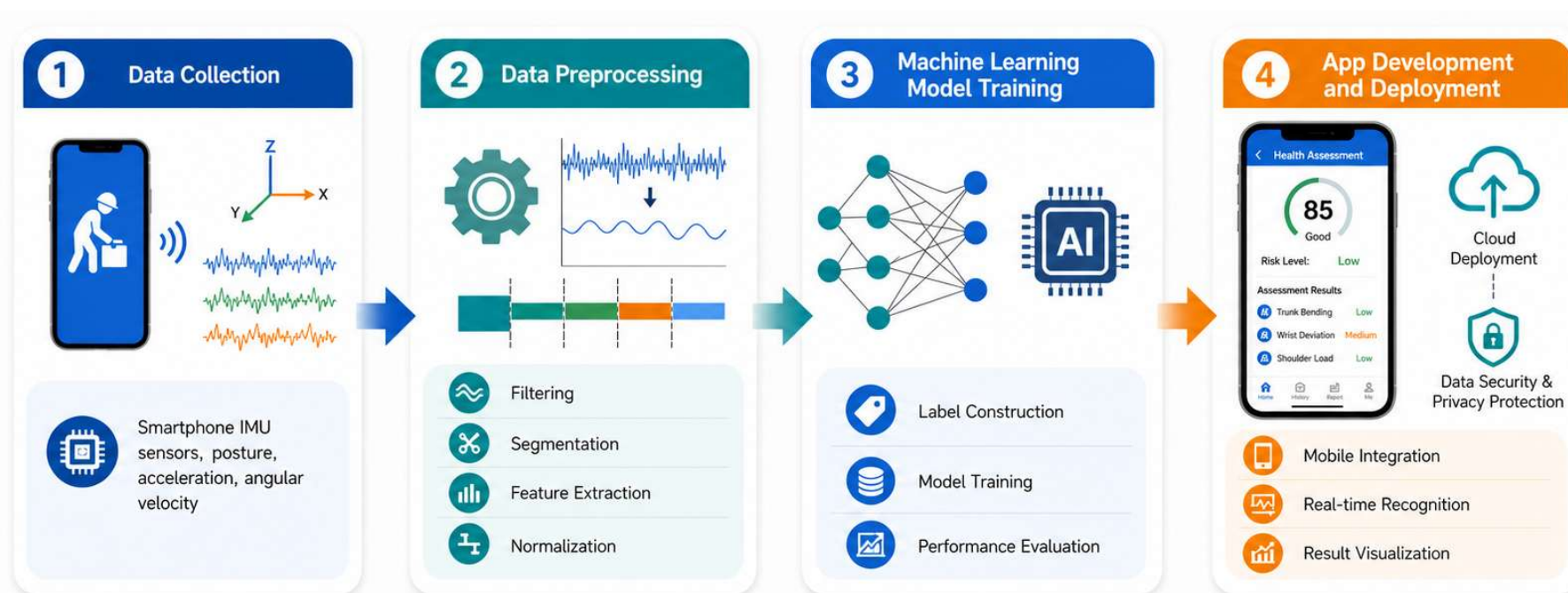
3. Our method – smartphone assessment

- While **traditional tools** provide valuable foundational frameworks, their inherent limitations—subjectivity, snapshot nature, and reliance on manual observation—have paved the way for a technological revolution in ergonomic assessment.
- A smartphone-based Low Back Disorder risk assessment app is introduced to achieve subjective assessment without external interference for workers.



3. Our method – smartphone assessment

- Demonstration using LBD assessment app



4. Coding Tutorial

A3 – ML training on posture classification

- Follow the instruction here:

https://colab.research.google.com/drive/1R5YcY5LDfWI6oIB45hPvSI2RHR_iHMFn?usp=sharing

- Evaluation Metrics: Accuracy, Precision, Recall, Macro F1-score

Accuracy	Predictions/ Classifications	$\frac{\text{Correct}}{\text{Correct} + \text{Incorrect}}$
Precision	Predictions/ Classifications	$\frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$
Recall	Predictions/ Classifications	$\frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$
F1	Predictions/ Classifications	$\frac{2 * \text{True Positive}}{\text{True Positive} + 0.5 (\text{False Positive} + \text{False Negative})}$



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THANK YOU!

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